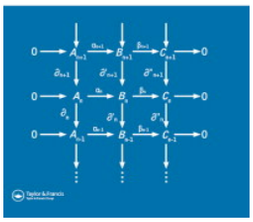

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Cohomology of pointed finite tensor categories

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ABSTRACT
We consider the finite generation property for cohomology algebra of pointed finite tensor categories via de-equivariantization and exact sequence of finite tensor categories. As a result, we prove that all coradically graded pointed finite tensor categories over abelian groups have finitely generated cohomology.

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1. Introduction

Let $\Bbbk$ be an algebraically closed field of characteristic zero. Given a finite tensor category $\mathcal{C}$ over $\Bbbk$, we may define the cohomology of $\mathcal{C}$ as follows:

$$\mathrm{H}^\bullet(\mathcal{C}, V) = \mathrm{Ext}^\bullet_{\mathcal{C}}(\mathbf{1}, V)$$

where V is an object in $\mathcal{C}$ and $\mathbf{1}$ is the unit object in $\mathcal{C}$. We mainly focus on cohomological finiteness property:

Definition 1.1. A finite tensor category $\mathcal{C}$ over $\Bbbk$ is said to have finitely generated cohomology (**FGC** for brevity), if $\mathrm{H}^\bullet(\mathcal{C}, \mathbf{1})$ is a finitely generated algebra and $\mathrm{H}^\bullet(\mathcal{C}, V)$ is a finitely generated $\mathrm{H}^\bullet(\mathcal{C}, \mathbf{1})$ -module for each V in $\mathcal{C}$.

In [10], Etingof and Ostrik proposed the well-known conjecture:

Conjecture 1.2. *Any finite tensor category over* $\Bbbk$ *satisfies* **FGC**.

For over twenty years, many beautiful results related to this conjecture have been obtained, especially from the perspective of Hopf algebras. For a finite-dimensional Hopf algebra H, then the category $\mathrm{Rep}(H)$ of finite-dimensional representations is a finite tensor category. By abuse of terminology, we say H satisfies **FGC**, if $\mathrm{Rep}(H)$ does. Ginzburg and Kumar in [13] showed that all small quantum groups satisfy **FGC**. Moreover, they computed their cohomology rings explicitly. In [16], Mastnak, Pevtsova, Schauenberg, and Witherspoon proved that any finite-dimensional pointed Hopf algebra over an abelian group (i.e., whose group-like elements form an abelian group) satisfies **FGC**, subject to mild restrictions on the group's order. This result was later extended to all finite-dimensional pointed Hopf algebras over abelian groups in [1]. Related results for fields of positive characteristic can be found in [18] and [11].

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On the other hand, some experts study this question from the perspective of category theory. In [17], Negron and Plavnik characterized when the dual category and the center of a finite tensor category $\mathcal{C}$ satisfy **FGC**, assuming $\mathcal{C}$ itself satisfies **FGC**. The condition of finitely generated cohomology is crucial in the theory of support varieties for finite tensor categories; further details may be found in [4] and [5].

We mainly focus on the coradically graded pointed finite tensor category $\mathcal{C}$ over abelian group. Recall that a finite tensor category is pointed if every simple object is invertible. In [2], the authors defined a finite tensor category as coradically graded if it is equivalent to the category of finite-dimensional comodules over a coradically graded coquasi-Hopf algebra. Then by the well-known Tannakian reconstruction theorem, there is a finite-dimensional coradically graded coquasi-Hopf algebra M over abelian group, such that $\mathcal{C} \cong \mathrm{Comod}(M)$. Thus the classification of coradically graded pointed finite tensor categories over abelian groups(up to tensor equivalence) is equivalent to the classification of finite-dimensional coradically graded coquasi-Hopf algebras over abelian groups(up to gauge equivalence).

The classification of such coquasi-Hopf algebras was completed in [14] and [15]. Based on the type of the associated twisted Nichols algebra, the classification divides into two cases: diagonal type and non-diagonal type. We address these cases separately to establish our main result on finitely generated cohomology:

Theorem 1.3. *Let $\mathcal{C}$ be a coradically graded pointed finite tensor category over abelian group, then $\mathcal{C}$ satisfies* **FGC**.

For convenience, we call a coradically graded coquasi-Hopf algebra of (non-)diagonal type if its corresponding twisted Nichols algebra is of (non-)diagonal type over the group of group-like elements.

We first consider coquasi-Hopf algebra M of diagonal type. The classification process is summarized in figures (4.3) and (4.17) in [14]. A key observation is that the figures exhibits a profound connection with the de-equivariantization of finite tensor categories, linking $\mathrm{Comod}(M)$ to the de-equivariantization of $\mathrm{Comod}(H)$, where H is a finite-dimensional coradically graded Hopf algebra of diagonal type. Using results from [17] and [1], we prove this case. It is worth emphasizing that Angiono and Galindo also proved similar conclusions on de-equivariantization of pointed finite tensor categories in [2], but our approach differs by employing a more algebraic perspective.

For a pointed coquasi-Hopf algebra M of non-diagonal type, a distinct method is necessary because its comodule category does not arise as a de-equivariantization of a pointed Hopf algebra comodule category. Instead, we analyze its intrinsic structure. By [15], $M \cong \mathcal{B}(V)\#\Bbbk G$ where $\mathcal{B}(V)$ is a twisted Nichols algebra of non-diagonal type, G is an abelian group and ω is a non-abelian cocycle. However, we may relate M to a pointed Hopf algebra of diagonal type via an exact sequence of tensor categories. Through further calculations and transformations, we demonstrate that $\mathrm{Comod}(M)$ satisfies **FGC** as well. thereby completing the proof of Theorem 1.3.

Section 2 introduces definitions, notations, and foundational results on coquasi-Hopf algebras and tensor categories. Section 3 treats the diagonal type case, while Section 4 addresses the non-diagonal type case and presents the proof of Theorem 1.3.

2. Background on coquasi-Hopf algebra and cohomology

This section reviews foundational definitions and results concerning coquasi-Hopf algebras and the cohomology of finite tensor categories.

2.1. Classification result of finite-dimensional coquasi-Hopf algebras

We briefly summarize the classification of finite-dimensional coradically graded pointed coquasi-Hopf algebras over finite abelian groups, as established in [15].

By definition, coquasi-Hopf algebras are exactly the dual of Drinfeld's quasi-Hopf algebras [7]. These are coassociative coalgebras that need not be associative. Analogous to the Hopf algebra case, the

classification relies critically on: twisted Yetter-Drinfeld module categories ${}^G_G\mathcal{YD}^\omega$ and twisted Nichols algebras, where G is a finite abelian group and ω is a normalized 3-cocycle over G. For formal definitions and examples, see [14 Section 2].

The classification of finite-dimensional coradically graded coquasi-Hopf algebra over abelian groups hinges on the structure of the coradical $(\Bbbk G, \omega)$, where ω is a 3-cocycle on G. A complete parametrization of 3-cohomology classes for finite abelian groups was achieved in [14] as follows. Let G be a finite abelian group and so there is no harm to assume that $G = Z_{m_1} \times Z_{m_2} \cdots \times Z_{m_n}$ with $m_i \mid m_{i+1}$ for $1 \le i \le n-1$. Denote $\mathscr{A}$ the set of all $\mathbb{N}$-sequences:

$$\underline{a} = (a_1, a_2, \ldots, a_l, \ldots, a_n, a_{12}, a_{13}, \ldots, a_{st}, \ldots, a_{n-1,n}, a_{123}, \ldots, a_{rst}, \ldots a_{n-2,n-1,n})$$

such that $0 \le a_l < m_l$, $0 \le a_{st} < (m_s, m_t)$, $0 \le a_{rst} < (m_r, m_s, m_t)$, with indices ordered lexicographically. Let g_i be the generator of Z_{m_i}, $1 \le i \le n$. For each $\underline{a} \in \mathscr{A}$, define

$$\omega\left(g_1^{i_1} \cdots g_n^{i_n}, g_1^{j_1} \cdots g_n^{j_n}, g_1^{k_1} \cdots g_n^{k_n}\right) = \prod_{l=1}^{n} \zeta_{m_l}^{a_l i_l \left[\frac{j_l + k_l}{m_l}\right]} \prod_{1 \le s < t \le n} \zeta_{m_s}^{a_{st} k_s \left[\frac{i_t + j_t}{m_t}\right]} \prod_{1 \le r < s < t \le n} \zeta_{(m_r, m_s, m_t)}^{a_{rst} k_r j_s i_t} \tag{2.1}$$

Here ζ_m denotes a primitive m-th root of unity. We further define $\mathscr{A}'$ as the subset of $\mathscr{A}$ satisfying $a_{rst} = 0$ for all $1 \le r < s < t \le n$ and $\mathscr{A}''$ as the subset of $\mathscr{A}$ satisfying $a_i = 0$, $a_{st} = 0$ for all $1 \le i \le n$, $1 \le s < t \le n$.

Lemma 2.1. [14, Proposition 3.8] *$\{\omega_{\underline{a}} | \underline{a} \in \mathscr{A}\}$ forms a complete set of representatives of the normalized 3-cocycles over G up to 3-cohomology.*

Among all the 3-cocycles over abelian groups, the class of abelian cocycles is particularly significant. These are intimately connected to the property that a twisted Yetter-Drinfeld module V over ${}^G_G\mathcal{YD}^\omega$ is of diagonal type, a crucial aspect of our classification. For details, see [14, Section 3] for details.

Lemma 2.2. [14, Corollary 3.13, Proposition 3.14] *Let G be a finite abelian group and ω a normalized 3-cocycle over G as in (2.1), then*

$$\begin{aligned} \omega \text{ is an abelian cocycle} &\iff a_{rst} = 0 \text{ for all } 1 \le r < s < t \le n \\ &\iff \text{Every Yetter-Drinfeld module over } (kG, \omega) \text{ is of diagonal type.} \end{aligned}$$

Abelian cocycles possess an additional remarkable property: they can be resolved within an extended abelian group. Consider $\mathbb{G} \cong Z_{m_1^2} \times Z_{m_2^2} \times \cdots \times Z_{m_n^2} = \langle \mathbb{g}_1 \rangle \times \langle \mathbb{g}_2 \rangle \times \cdots \times \langle \mathbb{g}_n \rangle$. There is a group surjection $\pi : \mathbb{G} \longrightarrow G$, $\mathbb{g}_i \mapsto g_i$, for $1 \le i \le n$. We may pullback the 3-cocycles over G and obtain a 3-cocycles over $\mathbb{G}$. That is, the map

$$\pi^*(\omega) : \mathbb{G} \times \mathbb{G} \times \mathbb{G} \longrightarrow \Bbbk^\times, \ \ (g, h, k) \mapsto \omega(\pi(g), \pi(h), \pi(k)).$$

is a 3-cocycle over $\mathbb{G}$. Actually, $\pi^*(\omega)$ is a coboundary. If we define $J_{\underline{a}}$ via

$$J_{\underline{a}} : \mathbb{G} \times \mathbb{G} \longrightarrow \Bbbk^\times, \ \ (\mathbb{g}_1^{x_1} \cdots \mathbb{g}_n^{x_n}, \mathbb{g}_1^{y_1} \cdots \mathbb{g}_n^{y_n}) \mapsto \prod_{l=1}^{n} \zeta_{m_l^2}^{a_l x_l (y_l - y_l')} \prod_{1 \le s < t \le n} \zeta_{m_s m_t}^{a_{st} x_t (y_s - y_s')},$$

then $\pi^*(\omega_{\underline{a}}) = \partial(J_{\underline{a}})$, This result corresponds precisely to [14].

The following theorem classifies finite-dimensional coradically graded pointed coquasi-Hopf algebras over finite abelian groups.

Theorem 2.3. [15, Theorem 5.2] *Let M be a finite-dimensional coradically graded pointed coquasi-Hopf algebra over a finite abelian group with the coradical $M_0 = (\Bbbk G, \omega)$. Then we have $M \cong \mathcal{B}(V) \# \Bbbk G$ for a Yetter-Drinfeld module of finite type $V \in {}^G_G\mathcal{YD}^\omega$.*

Remark 2.4. We elaborate on Theorem 2.3 and present related results. Recall that if $V \in {}^G_G\mathcal{YD}^\omega$, then we can define the support group $G_V :=< g \in G \mid^g V \neq 0 >$, which is a subgroup of G.

(i) A Yetter-Drinfeld module ${}^G_G\mathcal{YD}^\omega$ is said to be of finite type if

∘ V has a standard basis. This is equivalent to the condition that $\omega\mid_{G_V}$ is an abelian cocycle.

∘ $\Delta_{\chi,E}$ is an arithmetic root system and $\mathrm{ht}(\alpha) < \infty$ for all $\alpha \in \Delta_{\chi,E}$, where (χ, E) is a pair associated to V.

(ii) By [15, Theorem 3.9], $\mathcal{B}(V)$ is finite-dimensional if and only if V is finite type. Furthermore, in this case, $\mathcal{B}(V)$ is isomorphic to a Nichols algebra of diagonal type in ${}^{G_V}_{G_V}\mathcal{YD}^{\omega|_{G_V}}$. Conversely, if $\omega\mid_{G_V}$ is nonabelian, then $\mathcal{B}(V)$ is infinite-dimensional by [15, Theorem 3.13].

(iii) The classification divides into two cases:

∘ Finite-dimensional twisted Nichols algebras of diagonal type over (G, ω) with ω abelian;

∘ Finite-dimensional twisted Nichols algebras of non-diagonal type where $\omega\mid_{G_V}$ is abelian.

2.2. De-equivariantization of finite tensor categories

In this subsection, we review the definition of de-equivariantization and present key results related to this construction.

Definition 2.5. (i) Let $\mathcal{C}$ be a finite braided tensor category. Suppose there is a braided tensor functor $\widetilde{i} : \mathcal{C} \to Z(\mathcal{D})$, such that the functor $Q : \mathcal{C} \xrightarrow{\widetilde{i}} Z(\mathcal{D}) \xrightarrow{F} \mathcal{D}$ is a fully faithful tensor functor, where F is the forget functor. Then we say i is a central embedding.

(ii) For any central embedding $\mathrm{Rep}(G) \to \mathcal{D}$, we can define the de-equivariantization $\mathcal{D}_G$, which is the tensor category of $\mathcal{O}(G)$-modules in $\mathcal{D}$, where $\mathcal{O}(G)$ is the linear dual of the group algebra.

Let G be a finite group. There exists a profound relationship between braided central Hopf subalgebras of a Hopf algebra H and central embedding $\mathrm{Rep}(G) \to {}^H_H\mathcal{YD}$.

Definition 2.6. Let H be a finite-dimensional Hopf algebra. A braided central Hopf subalgebra of H is a pair (K, r), where $K \subset H$ is a Hopf subalgebra, and $r : H \otimes K \to k$ is a bilinear form such that:

$$r\left(hh', k\right) = r\left(h', k_1\right) r\left(h, k_2\right), \tag{2.2}$$

$$r\left(h, kk'\right) = r\left(h_1, k\right) r\left(h_2, k'\right), \tag{2.3}$$

$$r(h, 1) = \varepsilon(h), \quad r(1, k) = \varepsilon(k), \tag{2.4}$$

$$r\left(h_1, k_1\right) k_2 h_2 = h_1 k_1 r\left(h_2, k_2\right), \tag{2.5}$$

$$r\left(k, k'\right) = \varepsilon\left(kk'\right), \tag{2.6}$$

for all $k, k' \in K$, $h, h' \in H$.

Lemma 2.7. [3, Theorem 3.4] *Let H be a Hopf algebra and $K \subset H$ a commutative Hopf subalgebra. Then the following set of data are equivalent:*

(1) A map $r : H \otimes K \to k$ such that (K, r) is a braided central Hopf subalgebra of H.

(2) A braided tensor functor $F : {}^K\mathcal{M} \to \mathcal{Z}({}^H\mathcal{M}) = {}^H_H\mathcal{YD}$ such that the composition with the forgetful functor $\mathcal{Z}({}^H\mathcal{M}) = {}^H_H\mathcal{YD} \to {}^H\mathcal{M}$ is fully faithful.

2.3. Cohomology of finite tensor categories

In this section, $\mathcal{C}$ and $\mathcal{D}$ are finite tensor categories. We present essential lemmas regarding their cohomology for subsequent applications.

Lemma 2.8. [12, Theorem 1.2.1] *Let H be a finite-dimensional pointed Hopf algebra whose group of group-like elements is abelian. Then* $\mathrm{Comod}(H)$ *as well as* $\mathrm{Rep}(H)$ *satisfies* **FGC**.

Lemma 2.9. [17, Proposition 3.3] *If $F : \mathcal{D} \to \mathcal{C}$ is a surjective tensor functor, and $\mathcal{D}$ satisfies* **FGC**, then $\mathcal{C}$ satisfies **FGC**.

Lemma 2.10. [17, Theorem 4.4] *Suppose $F : \mathcal{D} \to \mathcal{C}$ is a de-equivariantization of $\mathcal{D}$ with respect to a central embedding* $\mathrm{Rep}(G) \to \mathcal{D}$, *or equivalently an equivariantization of $\mathcal{C}$ with respect to a G-action. Then $\mathcal{D}$ satisfies* **FGC** if and only if $\mathcal{C}$ satisfies **FGC**.

The authors of [8] generalized Bruguieres and Natale's definition of exact sequences of tensor categories, introducing a new notion of exact sequences with respect to module categories. For brevity, we omit the explicit definition here.

Lemma 2.11. [17, Corollary 8.13] *Let*

$$\mathcal{B} \longrightarrow \mathcal{C} \longrightarrow \mathcal{D} \boxtimes \mathrm{End}(\mathcal{M})$$

be an exact sequence of finite tensor categories. In particular, we assume $\mathcal{D}$ is a fusion category, then there is a natural identification of $\Bbbk$-algebras

$$\mathrm{H}^\bullet(\mathcal{B}, 1) = \mathrm{H}^\bullet(\mathcal{C}, 1).$$

Furthermore, for any object V in $\mathcal{C}$, there is an identification of $\mathrm{H}^\bullet(\mathcal{C}, 1)$-modules, $\mathrm{H}^\bullet(\mathcal{B}, \mathscr{H}^0_{\mathcal{C}}(\mathcal{D}, V)) = \mathrm{H}^\bullet(\mathcal{C}, V)$, where the definition of $\mathscr{H}^0_{\mathcal{C}}$ is given in [17, Definition 8.10].

3. On finite-dimensional coquasi-Hopf algebras of diagonal type

In this section, given a finite-dimensional coradically graded coquasi-Hopf algebra of diagonal type, we prove that $\mathrm{Comod}(M)$ satisfies **FGC**. We begin by recalling key classification results.

3.1. Overview of classification results

In this section, let $G \cong Z_{m_1} \times Z_{m_2} \times \cdots \times Z_{m_n} = \langle g_1 \rangle \times \langle g_2 \rangle \times \cdots \times \langle g_n \rangle$ with an abelian 3-cocycle $\omega_{\underline{a}}$ and $\mathbb{G} \cong Z_{m_1^2} \times Z_{m_2^2} \times \cdots \times Z_{m_n^2} = \langle \mathbb{g}_1 \rangle \times \langle \mathbb{g}_2 \rangle \times \cdots \times \langle \mathbb{g}_n \rangle$. We assume $\mathcal{B}(V) \in {}^G_G\mathcal{YD}^{\omega_{\underline{a}}}$ is a finite-dimensional Nichols algebra of diagonal type. Without loss of generality, we can assume $G = G_V$ (the support group of V). As detailed in [14 Section 4], the authors provided a method to classify such twisted Nichols algebras. Here, we adopt an equivalent approach to describe the 'return trip' transformation between twisted and ordinary Nichols algebras for later use.

$$\begin{tikzcd}
\text{Given } \mathcal{B}(V) \in {}^G_G\mathcal{YD}^{\omega_{\underline{a}}} \arrow[d] \\
\mathcal{B}(V) \cong \mathcal{B}(V_{\mathcal{D}}) \in {}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\omega_{\underline{a}})} \arrow[d] \\
\mathcal{B}(V_{\mathcal{D}}) \text{ is twisted equivalent to } \mathcal{B}(V_{\mathcal{D}})^{J_{\underline{a}}^{-1}} \in {}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}
\end{tikzcd}$$

Figure I

$$\begin{tikzcd}
\mathcal{B}(V')^{J_{\underline{a}}} \cong \mathcal{B}(V) \in {}^G_G\mathcal{YD}^{\omega_{\underline{a}}} \\
\mathcal{B}(V')^{J_{\underline{a}}} \in {}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}^{\partial(J_{\underline{a}})} \arrow[u, "\text{If [14, (4.8)] holds}"'] \\
\text{Given Nichols algebra } \mathcal{B}(V') \in {}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD} \arrow[u]
\end{tikzcd}$$

Figure II

Let us illustrate the above figures explicitly via the language of root data. Let $\mathcal{D} = (\mathcal{D}_{\chi,E}, S, X)$ be a root data, where:

∘ $\mathcal{D}_{\chi,E}$ is a Dynkin diagram of an arithmetic root system $\Delta_{\chi,E}$;
∘ $S=(s_{ij})_{m\times n}, X=(x_{ij})_{n\times m}$ are two matrices satisfying certain conditions (see [14, Definition 4.14]).

By [14, Corollary 4.13], for each root data $\mathcal{D}$, there exists unique $\underline{a}\in\mathscr{A}'$ such that [14, (4.8)] holds for all $1\le i\le n$ and $1\le j\le m$.

$$\zeta_{m_i^2}^{m_ix_{ij}}=\zeta_{m_i^2}^{a_is_{ji}m_i}\prod_{i<k\le n}\zeta_{m_im_k}^{a_{ik}s_{jk}m_i} \tag{3.1}$$

For each root data $\mathcal{D}$, we can define a Nichols algebra $\mathcal{B}(V_{\mathcal{D}})\in{}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\omega_{\underline{a}})}$ as follows. Let $V_{\mathcal{D}}$ be the Yetter-Drinfeld module in ${}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\omega_{\underline{a}})}$ with a canonical basis $\{X_i\mid 1\le i\le m\}$ such that

$$\delta_L(X_i)=\prod_{k=1}^{n}\mathbb{g}_k^{s_{ik}}\otimes X_i,\quad \mathbb{g}_i\triangleright X_j=\zeta_{m_i^2}^{x_{ij}}\frac{J_{\underline{a}}\left(g_i,\prod_{k=1}^{n}\mathbb{g}_k^{s_{ik}}\right)}{J_{\underline{a}}\left(\prod_{k=1}^{n}\mathbb{g}_k^{s_{ik}},g_i\right)}X_j.$$

The second statement of [14, Theorem 4.15] tells us that each twisted Nichols algebra in ${}^{G}_{G}\mathcal{YD}^{\omega_{\underline{a}}}$ must be isomorphic to $\mathcal{B}(V_{\mathcal{D}})\in{}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\omega_{\underline{a}})}$ for some root data $\mathcal{D}$. This is exactly what Figure I says.

The first part of that theorem shows that each Nichols algebra in ${}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\omega_{\underline{a}})}$ satisfying equation (3.1) must be isomorphic to $\mathcal{B}(V_{\mathcal{D}})\in{}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\omega_{\underline{a}})}$ for some root data $\mathcal{D}$. Moreover, $\mathcal{B}(V_{\mathcal{D}})$ will be uniquely isomorphic to a twisted Nichols algebra in ${}^{G}_{G}\mathcal{YD}^{\omega_{\underline{a}}}$. This corresponds to the Figure II.

This completes the classification of finite-dimensional twisted Nichols algebras of diagonal type, and consequently, all finite-dimensional coquasi-Hopf algebras of diagonal type.

3.2. Proof of diagonal type case

For each $V'\in{}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\omega_{\underline{a}})}$, one may define a finite-dimensional Nichols algebra $\mathcal{B}(V')\in{}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\omega_{\underline{a}})}$ and thus a finite-dimensional coquasi-Hopf algebra $H^{J_{\underline{a}}}=\mathcal{B}(V')\#\Bbbk\mathbb{G}$. Through the processes of twisting and bosonization, we obtain a finite-dimensional coradically graded pointed Hopf algebra:

$$H=\mathcal{B}(V')^{J_{\underline{a}}^{-1}}\#\Bbbk\mathbb{G}.$$

On the other hand, if $\underline{a}$ satisfies equation (3.1), we can define a finite-dimensional coquasi-Hopf algebra

$$M\cong\mathcal{B}(V)\#\Bbbk G,$$

where $\mathcal{B}(V)$ is isomorphic to $\mathcal{B}(V')$. Note that $\mathbb{G}$ contains a subgroup $\widetilde{\mathbb{G}}=\langle\mathbb{g}_1^{m_1}\rangle\times\langle\mathbb{g}_2^{m_2}\rangle\times\cdots\times\langle\mathbb{g}_n^{m_n}\rangle\cong G$. We now state an observation:

Proposition 3.1. *With the above notions, if equation (3.1) holds for all $1\le i\le n$ and $1\le j\le m$, then we have such a tensor equivalence*

$$\mathrm{Comod}(H)_{\widetilde{\mathbb{G}}}\cong\mathrm{Comod}(M). \tag{3.2}$$

Proof. Let χ_i be the generator of $\widehat{Z_{m_i^2}}$ such that $\chi_i(\mathbb{g}_i)=\zeta_{m_i^2}$ for all $1\le i\le n$. Define $\phi:\widetilde{\mathbb{G}}\to\widehat{\mathbb{G}}$ by:

$$\phi((\mathbb{g}_1^{m_1})^{y_1}\cdots(\mathbb{g}_n^{m_n})^{y_n})=\prod_{l=1}^{n}\chi_l^{a_ly_lm_l}\prod_{1\le j<k\le n}\chi_k^{a_{jk}y_jm_k}. \tag{3.3}$$

Let $k=(\mathbb{g}_1^{m_1})^{y_1}\cdots(\mathbb{g}_n^{m_n})^{y_n}\in\widetilde{\mathbb{G}}$. Then for any $k'\in\widetilde{\mathbb{G}}$ and each generator X_j, $1\le j\le m$, we have

$$<k',\phi(k)>=1,$$

$$k\triangleright X_j=\prod_{i=1}^{n}\zeta_{m_i^2}^{m_iy_ix_{ij}}X_j\overset{(3.1)}{=}\prod_{i=1}^{n}\zeta_{m_i^2}^{a_iy_is_{ji}m_i}\prod_{1\le i<l\le n}\zeta_{m_im_l}^{a_{il}s_{jl}y_im_i}X_j=<\prod_{l=1}^{n}\mathbb{g}_l^{s_{jl}},\phi(k)>X_j.$$

Hence, the pair $(\widetilde{\mathbb{G}}, \phi)$ satisfies the condition (b) in [3, Proposition 4.3], yielding a braided central Hopf subalgebra $(\Bbbk\widetilde{\mathbb{G}}, r)$ of H. As $\widetilde{\mathbb{G}}$ is abelian, we have $\mathrm{Rep}(\widetilde{\mathbb{G}}) \cong \mathrm{Comod}(\Bbbk\widetilde{\mathbb{G}})$ as fusion category. By Lemma 2.7, there exists a braided tensor functor $F : \mathrm{Rep}(\widetilde{\mathbb{G}}) \to \mathcal{Z}({}^H\mathcal{M}) = {}^H_H\mathcal{YD}$ whose composition with the forgetful functor $\mathcal{Z}({}^H\mathcal{M}) = {}^H_H\mathcal{YD} \to {}^H\mathcal{M}$ is fully faithful. This guarantees the existence of the de-equivariantization $\mathrm{Comod}(H)$, which defined as the tensor category of left $\Bbbk\widetilde{\mathbb{G}}$-modules in $\mathrm{Comod}(H)$.

On the other hand, there is an epimorphism of coquasi-Hopf algebras:

$$\pi := (f \otimes p) : H^{J_{\underline{a}}} = \mathcal{B}(V_{\mathcal{D}})\#\Bbbk\mathbb{G} \longrightarrow M = \mathcal{B}(V)\#\Bbbk G.$$

where $p : \Bbbk\mathbb{G} \to \Bbbk G$ is given by $\mathbb{g}_i \mapsto g_i$ is surjective and $f : \mathcal{B}(V_{\mathcal{D}}) \to \mathcal{B}(V)$ is an isomorphism. Direct computation shows:

$$\Bbbk\widetilde{G} = (H^{J_{\underline{a}}})^{\mathrm{co}\,\pi} := \left\{x \in H^{J_{\underline{a}}} \mid (\mathrm{id} \otimes \pi)(\Delta(b)) = b \otimes 1\right\}.$$

It follows by [6, Example 6.4] that $\Bbbk\widetilde{\mathbb{G}}$ admits a structure of commutative algebra in $\mathcal{Z}(\mathrm{Comod}(H^{J_{\underline{a}}}))$, making the tensor category of left $\Bbbk\widetilde{\mathbb{G}}$-modules in $\mathrm{Comod}(H^{J_{\underline{a}}})$ is tensor equivalent to $\mathrm{Comod}(M)$. Note that, $\mathrm{Comod}(H) \cong \mathrm{Comod}(H^{J_{\underline{a}}})$ as tensor categories since $H^{J_{\underline{a}}}$ and H differs by a cocycle-deformation. Hence the tensor category of left $\Bbbk\widetilde{\mathbb{G}}$-modules in $\mathrm{Comod}(H)$ is tensor equivalent to $\mathrm{Comod}(M)$. We conclude: $\mathrm{Comod}(H)_{\widetilde{\mathbb{G}}} \cong \mathrm{Comod}(M)$. □

Theorem 3.2. *For each finite-dimensional coradically graded coquasi-Hopf algebra M of diagonal type,* $\mathrm{Comod}(M)$ *satisfies* **FGC**.

Proof. Given a finite-dimensional coradically graded pointed coquasi-Hopf algebra $M \cong \mathcal{B}(V)\#\Bbbk G$ of diagonal type, Figure I associates it to a unique Hopf algebra $H \cong \mathcal{B}(V_{\mathcal{D}})^{J_{\underline{a}}^{-1}}\#\Bbbk\mathbb{G}$. The equation (3.1) holds automatically for $\mathcal{B}(V_{\mathcal{D}})$ by [14, Theorem 4.15]. Then by Proposition 3.1, there is a subgroup $\widetilde{\mathbb{G}} \subseteq \mathbb{G}$ such that $\mathrm{Comod}(H)_{\widetilde{\mathbb{G}}} \cong \mathrm{Comod}(M)$. Since $\mathrm{Comod}(H)$ satisfies **FGC** by Lemma 2.9, so does $\mathrm{Comod}(M)$ by Lemma 2.10. □

Remark 3.3. The referee pointed out that [3, Proposition 4.3] can simplify the proof of Proposition 3.1. We acknowledge referee for providing this better approach.

4. Non-diagonal type case and proof of Theorem 1.3

The classification result of finite-dimensional twisted Nichols algebras of non-diagonal type is much simpler, but the procedure is more difficult, see [15]. We first characterize all finite-dimensional simple twisted Nichols algebras of non-diagonal type over abelian groups.

Lemma 4.1. [15, Proposition 2.12] *Suppose $V \in {}^G_G\mathcal{YD}^{\omega}$ is a simple twisted Yetter-Drinfeld module of non-diagonal type,* $\deg(V) = g$. *Then $\mathcal{B}(V)$ is finite-dimensional if and only if V is one of the following two cases:*

(C1) $g \triangleright v = -v$ *for all* $v \in V$,

(C2) $\dim(V) = 2$ *and* $g \triangleright v = \zeta_3 v$ *for all* $v \in V$.

Remark 4.2. In [15], the authors showed the following results. (1) For a rank-2 Yetter-Drinfeld module $V = V_1 \oplus V_2$ with V_1, V_2 are simple but nonisomorphic, $\mathcal{B}(V)$ is finite-dimensional if and only if $\dim(V_1) = \dim(V_2) = 2$ and V_1, V_2 are of type (C1) with some additional restriction.

(2) Any V of rank 3 with pairwise nonisomorphic simple twisted Yetter-Drinfeld modules yields an infinite-dimensional Nichols algebra $\mathcal{B}(V)$.

For brevity, we focus on simple twisted Yetter-Drinfeld modules of non-diagonal type, as our methods extend directly to rank 2. Consider the group

$$G \cong \langle g_1, g_2, \ldots, g_n \rangle$$

for $n \geq 3$, since ω is a non-abelian 3-cocycle. Let $V \in {}^G_G\mathcal{YD}^\omega$ be of either type (C1) or (C2) with $\deg(V) = g_1$ and $\dim(V) = m \geq 2$. By [15, Proposition 3.10, Theorem 3.9], there is a finite abelian group

$$\mathbb{G} \cong \langle \mathrm{g}_1 \rangle \times \langle \mathrm{g}_2 \rangle \times \cdots \times \langle \mathrm{g}_n \rangle$$

with generators g_i, such that $\mathrm{ord}(\mathrm{g}_i) = \mathrm{ord}(g_i)^2$. Moreover, there exists a group epimorphism $\pi : \mathbb{G} \longrightarrow G$ satisfying $\pi(\mathrm{g}_i) = g_i$ and a section ι to π such that $\iota(g_i) = \mathrm{g}_i$ for all $1 \leq i \leq n$. Furthermore, there exists $\widetilde{V} \in {}^{\mathbb{G}}_{\mathbb{G}}\mathcal{YD}^{\pi^*\omega}$ such that $\mathcal{B}(\widetilde{V}) \cong \mathcal{B}(V)$, $\pi^*(\omega) \in \mathscr{A}''$ and the support group $\mathbb{G}_{\widetilde{V}}$ of $\widetilde{V}$ is $\langle \mathrm{g}_1 \rangle$. The following lemma is immediate.

Lemma 4.3. *The bosonization $H := \mathcal{B}(\widetilde{V})\#\Bbbk\mathbb{G}_{\widetilde{V}}$ is a Hopf algebra and satisfies* **FGC**.

Proof. Since $\pi^*\omega \in \mathscr{A}''$, the restriction $\pi^*\omega \mid_{\mathbb{G}_{\widetilde{V}} \times \mathbb{G}_{\widetilde{V}} \times \mathbb{G}_{\widetilde{V}}} \equiv 1$. By [15, Corollary 3.16], $\mathcal{B}(\widetilde{V})$ is isomorphic to a Nichols algebra of diagonal type in ${}^{\mathbb{G}_{\widetilde{V}}}_{\mathbb{G}_{\widetilde{V}}}\mathcal{YD}$(retaining the notation $\mathcal{B}(\widetilde{V})$ for simplicity). Thus $H := \mathcal{B}(\widetilde{V})\#\Bbbk\mathbb{G}_{\widetilde{V}}$ is a Hopf algebra, with an abelian group of group-like elements. Hence H satisfies **FGC** by Lemma 2.8. □

Let $M := \mathcal{B}(V)\#\Bbbk G$ and $\widetilde{M} := \mathcal{B}(\widetilde{V})\#\Bbbk\mathbb{G}$. To prove $\mathrm{Comod}(M)$ satisfies **FGC**, we are going to show $\mathrm{Comod}(\widetilde{M})$ does, leveraging the surjective coquasi-Hopf algebra map $\widetilde{M} \to M$.

Theorem 4.4. *Both* $\mathrm{Comod}(\widetilde{M})$ *and* $\mathrm{Comod}(M)$ *satisfy* **FGC**.

Proof. Clearly, $\widetilde{M}$ and H are generated by group-like and skew primitive elements. Since $\dim(V) = m$ and $\mathcal{B}(V) \cong \mathcal{B}(\widetilde{V})$. We may assume $\widetilde{V} = \mathrm{span}\,\{x_1, x_2, \ldots, x_m\}$. Thus, $\widetilde{M}$ is generated by

$$\left\{ x_1, x_2, \ldots, x_m, \mathrm{g}_j^i \mid 0 \leq i < m_j^2,\ 1 \leq j \leq n \right\},$$

while H is generated by

$$\left\{ x_1, x_2, \ldots, x_m, \mathrm{g}_1^i \mid 0 \leq i < m_1^2 \right\}.$$

The injective coquasi-Hopf algebra map $i : H \to \widetilde{M}$, $x_j\#\mathrm{g}_1^i \mapsto x_j\#\mathrm{g}_1^i$ for $0 \leq i \leq n-1$, $1 \leq j \leq m$, induces a fully faithful tensor functor

$$\iota : \mathrm{Comod}(H) \longrightarrow \mathrm{Comod}(\widetilde{M}). \tag{4.1}$$

Note that $p : \mathbb{G} \to \mathbb{G}/\mathbb{G}_{\widetilde{V}}$ is a group epimorphism given by $p(\mathrm{g}_1) = 1$ and $p(\mathrm{g}_i) = \mathrm{g}_i$ for all $2 \leq i \leq n$ and $\pi : \mathcal{B}(V) \to \Bbbk$, $1_{\mathcal{B}(\mathcal{V})} \mapsto 1_{\mathcal{B}(V)}$, $x_i \mapsto 0$ is an algebra map and a coalgebra map. This yields a coquasi-Hopf algebra surjection:

$$f = (\pi \otimes p) : \mathcal{B}(\widetilde{V})\#\Bbbk\mathbb{G} \longrightarrow \Bbbk\mathbb{G}/\mathbb{G}_{\widetilde{V}},\ x\#\mathrm{g} \mapsto \pi(x)\#p(\mathrm{g}).$$

which induces a surjective tensor functor:

$$F : \mathrm{Comod}(\widetilde{M}) \to \mathrm{Vec}^{\omega_{\mathbb{G}/\mathbb{G}_{\widetilde{V}}}}_{\mathbb{G}/\mathbb{G}_{\widetilde{V}}}. \tag{4.2}$$

We denote the fusion category $\mathrm{Vec}^{\omega_{\mathbb{G}/\mathbb{G}_{\widetilde{V}}}}_{\mathbb{G}/\mathbb{G}_{\widetilde{V}}}$ as $\mathcal{D}$ for simplicity. Since $f \circ i(x\#\mathrm{g}_1^i) = \pi(x)\#1 \in \Bbbk$, we see that $F \circ \iota(X) \in \mathrm{Vec}$ for all $X \in \mathrm{Comod}(H)$ by definition of ι and F. That is $\mathrm{Comod}(H) \subseteq \mathrm{Ker}(F)$.

All categories involved are comodule categories of finite-dimensional coquasi-Hopf algebras. Hence $\mathrm{FPdim}(\mathrm{Comod}(H)) = \dim_{\Bbbk}(H)$, $\mathrm{FPdim}(\mathrm{Comod}(\widetilde{M})) = \dim_{\Bbbk}(\widetilde{M})$ and $\mathrm{FPdim}(\mathcal{D}) = \dim_{\Bbbk} \Bbbk(\mathbb{G}/\mathbb{G}_{\widetilde{V}})$ by [9, Example 6.1.9]. Clearly,

$$\dim_{\Bbbk}(\widetilde{M}) = \dim_{\Bbbk}(H) \cdot \dim_{\Bbbk} \Bbbk(\mathbb{G}/\mathbb{G}_{\widetilde{V}}).$$

Hence

$$\mathrm{FPdim}(\mathrm{Comod}(\widetilde{M})) = \mathrm{FPdim}(\mathrm{Comod}(H)) \cdot \mathrm{FPdim}(\mathcal{D}). \tag{4.3}$$

By [8, Theorem 3.4], there is an exact sequence of finite tensor category

$$\mathrm{Comod}(H) \longrightarrow \mathrm{Comod}(\widetilde{M}) \longrightarrow \mathcal{D}. \tag{4.4}$$

Lemma 2.11 provides a natural identification of $\Bbbk$-algebras

$$\mathrm{H}^{\bullet}(\mathrm{Comod}(H), 1) = \mathrm{H}^{\bullet}(\mathrm{Comod}(\widetilde{M}), 1).$$

Since $\mathrm{Comod}(H)$ satisfies **FGC** by Lemma 4.3, $\mathrm{H}^{\bullet}(\mathrm{Comod}(\widetilde{M}), 1)$ is a finitely generated algebra. Moreover, for each $W \in \mathrm{Comod}(\widetilde{M})$, there is an identification of $\mathrm{H}^{\bullet}(\mathrm{Comod}(\widetilde{M}), 1)$ module via

$$\mathrm{H}^{\bullet}(\mathrm{Comod}(H), \mathscr{H}^0_{\mathrm{Comod}(\widetilde{M})}(\mathcal{D}, W)) = \mathrm{H}^{\bullet}(\mathrm{Comod}(\widetilde{M}), W).$$

Since $\mathrm{H}^{\bullet}(\mathrm{Comod}(H), \mathscr{H}^0_{\mathrm{Comod}(\widetilde{M})}(\mathcal{D}, W))$ is a finitely generated module of $\mathrm{H}^{\bullet}(\mathrm{Comod}(H), 1)$, then $\mathrm{H}^{\bullet}(\mathrm{Comod}(\widetilde{M}), W)$ is a finitely generated module of $\mathrm{H}^{\bullet}(\mathrm{Comod}(\widetilde{M}), 1)$. Hence $\mathrm{Comod}(\widetilde{M})$ satisfies **FGC**. Recall that there is a surjection of coquasi-Hopf algebra $\widetilde{M} \to M$, which induces a surjective tensor functor $\mathrm{Comod}(\widetilde{M}) \to \mathrm{Comod}(M)$. Thus $\mathrm{Comod}(M)$ satisfies **FGC** by Lemma 2.9. □

Proof of Theorem 1.3. Since $\mathcal{C} \cong \mathrm{Comod}(M)$ for some finite-dimensional coradically graded coquasi-Hopf algebra M over an abelian group, the result follows directly from Theorems 3.2 and 4.4. □

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